


BOOLEAN - ALGEBRA

Laws & Properties

Boolean Algebra - Minimization

└ Closure; A, B ins

└ $A + B \in S$ } and 0 or 1 or min/max T, F

└ $AB \in S$

└ Commutative; $AB = BA$

└ $A + B = B + A$

└ $AB = BA$

Self-Operation

└ $A + A = A$

└ $A \cdot A = A$

└ Associative; $ABC = CAB$

└ $A + (B + C) = (A + B) + C$

└ $A(BC) = (AB)C$

└ Distributive; $AB + AC = A(B + C)$

└ $A + BC = (A + B)(A + C)$

└ $AC(B + C) = AB + AC$

└ Identities

└ $A + 0 = A$ } OR

└ $A \cdot 1 = A$ } AND

└ $A + 1 = 1$ }

└ $A \cdot 0 = 0$ }

└ Complement

└ $\overline{(\overline{A})} = A$

└ $A + \overline{A} = 1$

└ $A\overline{A} = 0$

Useful laws

└ $AB + A\overline{B} = A$

└ $A + AB = A$

└ $A + \overline{A}B = A + B$

└ De Morgan's

└ $\overline{A + B} = \overline{A}\overline{B}$

└ $\overline{AB} = \overline{A} + \overline{B}$

Summary

All the highlight, those are ESSENTIAL. KEEP IT IN UR MINE

Minimization Exercises 1

Cue

အကယ်၍ ခုနစ်နှစ်က နေထိုင်ခဲ့သည့် နေရာကို ပြန်လည် ခြေထောက်ထားရပါက နေရာအသစ်ကို ရွေးချယ်ရာတွင် နေရာအသစ်ကို ရွေးချယ်ရာတွင် နေရာအသစ်ကို ရွေးချယ်ရာတွင်

⑦ $Y = BC + \bar{A}BC + ABC$

$$= B\bar{C} + BC(\bar{A}+A)$$

$$= B\bar{C} + BC$$

$$= BC \bar{C} + C$$

= B — #

Hint

$$0 \cdot 0 \cdot 0 = 0$$

$$0 \ 0 \ 1 = 1$$

$$0 \mid 0 = 2$$

$$0 \quad 1 \quad 1 \quad = 3$$

$$1 \cup 0 = 1$$

10125

1 1 0 2 0

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$$1000 = 8$$

1 0 0 1 = 9

1 0 1 0 $\approx 10, A$

1 0 1 1 2 11, B

$$1 \quad 1 \quad 0 \quad 0 \quad : 12, C$$

$$1101 = 13,0$$

$$110 = 14, E$$

$$11 \quad | \quad f = 15, f$$

Binary	Decimal	Hex
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F